Equation of line passing through 1 point

The slope intercept form of a straight line is used to obtain a line's equation. The slope of the line and the intercept cut by the line with the y-axis are required for the slope intercept formula.

It is given by

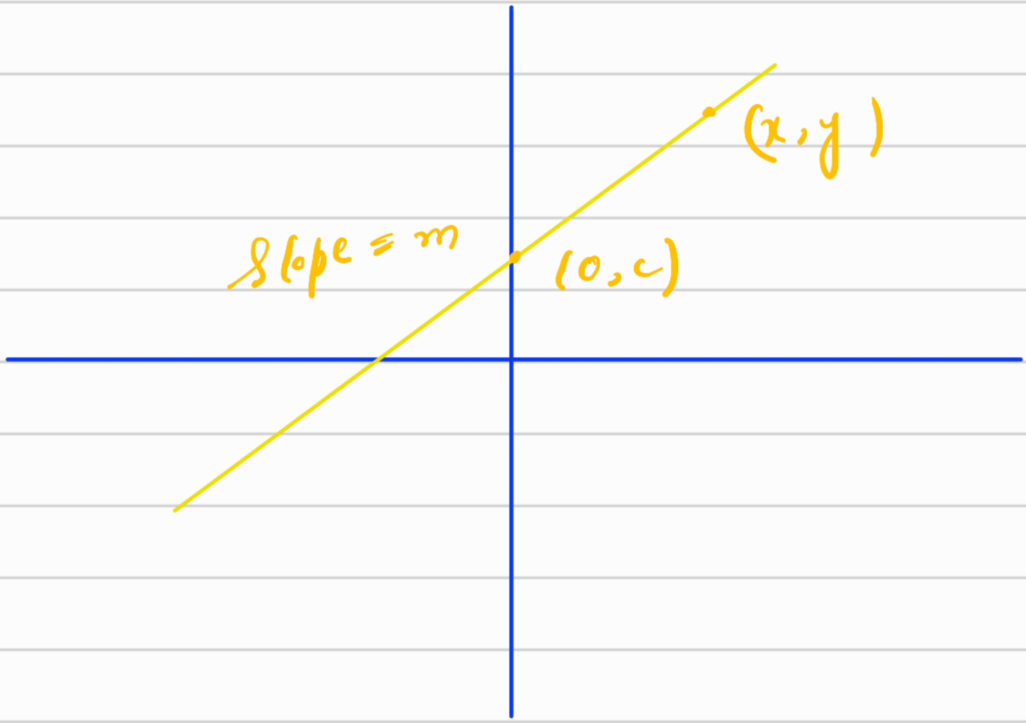
$$\boxed{y = mx + c}$$

where

m = slope of the line

c = y-intercept of the line.

Derivation



Consider 2 points on the above line $(x_1, y_1) = (0, c)$ and $(x_2, y_2) = (x, y)$

The slope of the line is given by,

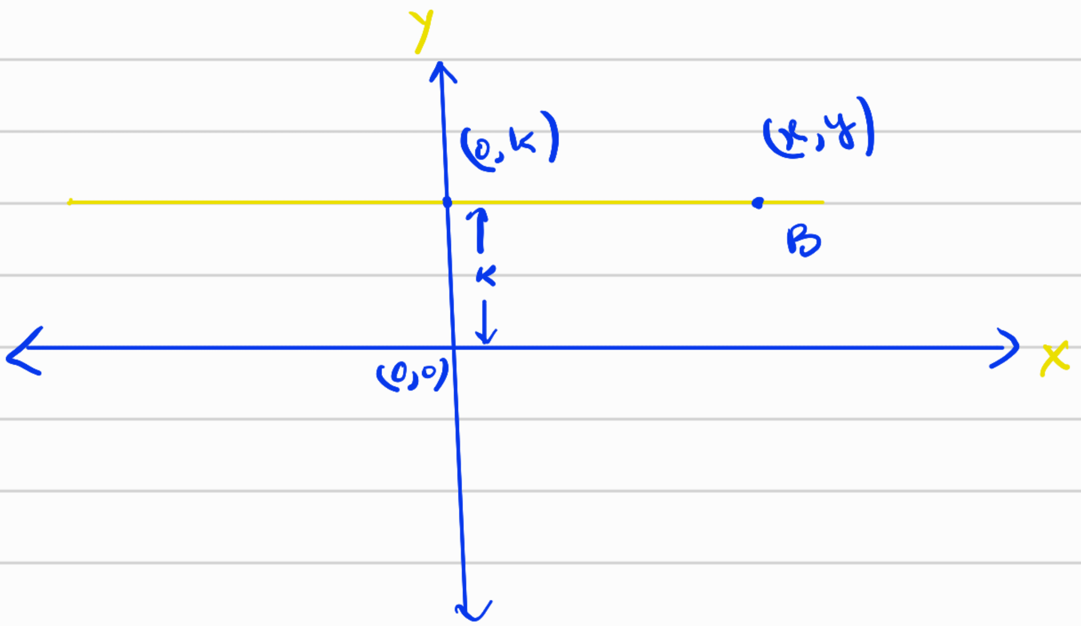
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\Rightarrow m = \frac{y - c}{x - 0}$$

$$\Rightarrow mx = y - c$$

$$\Rightarrow \boxed{y = mx + c}$$

Equation of line parallel to x axis



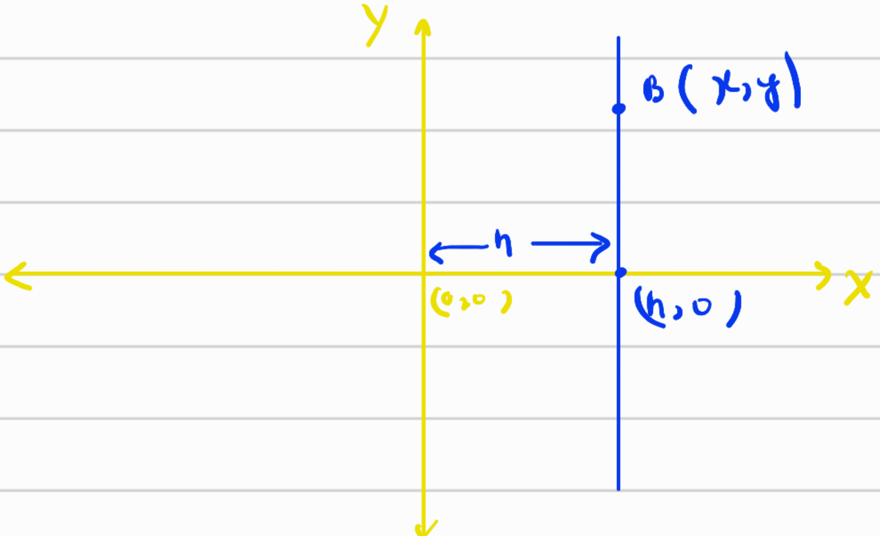
Let the distance of the line for x axis be k

∴ the y-coordinate will be always k as the line is parallel to x-axis

lets take a point B on the line with coordinate = (x, y)
we know that $y = k$

$\therefore$ Equation = $\boxed{y = k}$

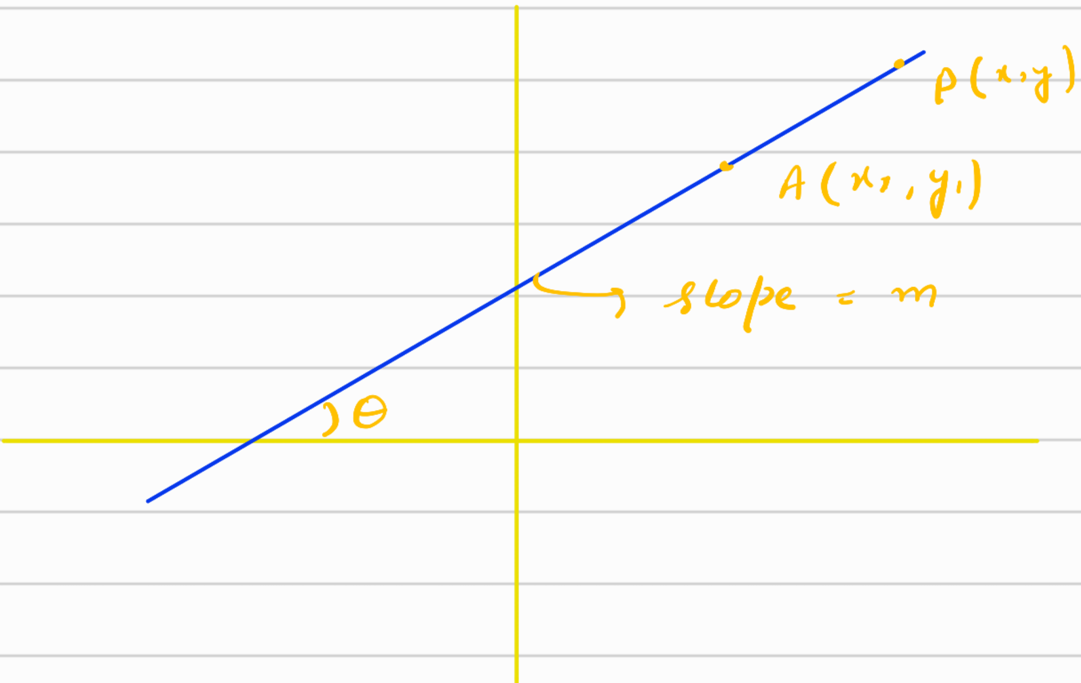
Equation of line parallel to y-axis



lets take variable point $B(x,y)$.
as we can see that x coordinate is remaining constant. i.e, h from the y-axis

$\therefore$ $\boxed{x = h}$

Slope - point form / when we have been given one point on the line



Lets us take a variable pt. $P(x,y)$ on the line

$\therefore$ slope of $AP = \dfrac{y-y_1}{x-x_1} = m$

$$\Rightarrow \boxed{y - y_1 = m(x - x_1)}$$

NOTE :-
To get the slope between two points we use

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Example on Slope-point form

We have been given a point $A(1,2)$ and slope $(m) = 2$ of a line. Find the equation.

Sol: Here One point is given $A(1,2)$ and slope $(m) = 2$.

$\therefore$ we use slope-point form.

$$y - y_1 = m(x - x_1)$$

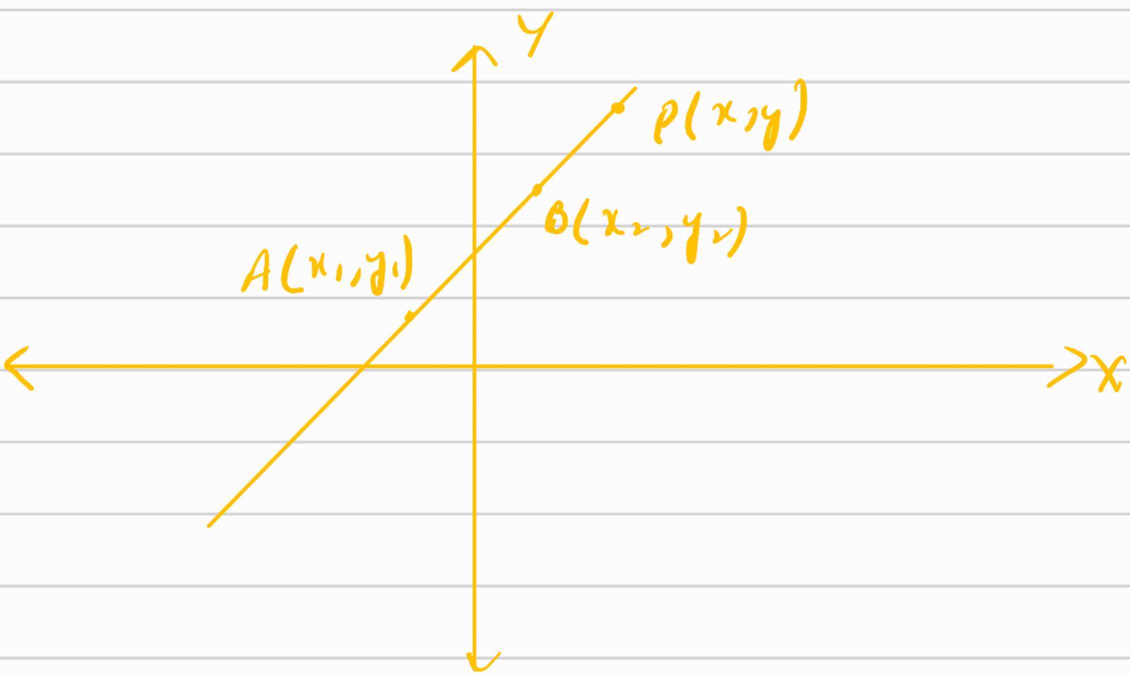
$$\Rightarrow y - 2 = 2(x - 1)$$

$$\Rightarrow y - \cancel{2} = 2x - \cancel{2}$$

$$\Rightarrow y = 2x$$

$$\Rightarrow \boxed{y - 2x = 0}$$

When two points are given



lets take a variable point $P(x,y)$ on the line.

Now,

slope of $AB = \dfrac{y_2 - y_1}{x_2 - x_1}$

slope of $AP = \dfrac{y - y_1}{x - x_1}$

slope of AB = slope of AP because both are of same line.

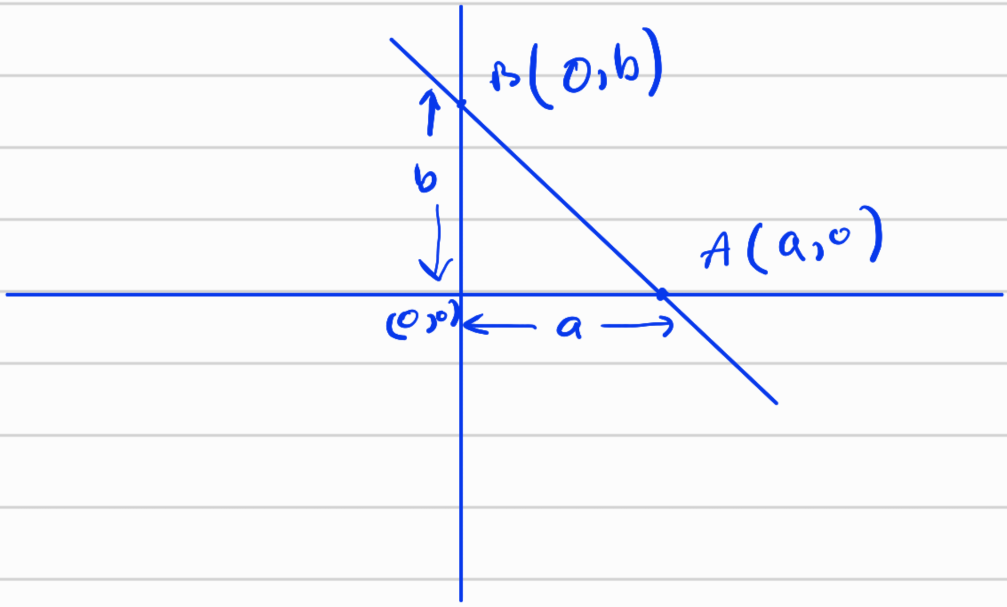
$\therefore \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{y - y_1}{x - x_1}$

$\Rightarrow \dfrac{y - y_1}{x - x_1} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$\Rightarrow \boxed{y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)}$

Here slope$(m) = \dfrac{y_2 - y_1}{x_2 - x_1}$

Intercept form



Slope of AB $= \dfrac{b-0}{0-a} = -\dfrac{b}{a} = m$

Now we know that

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 0 = -\frac{b}{a}(x - a)$$

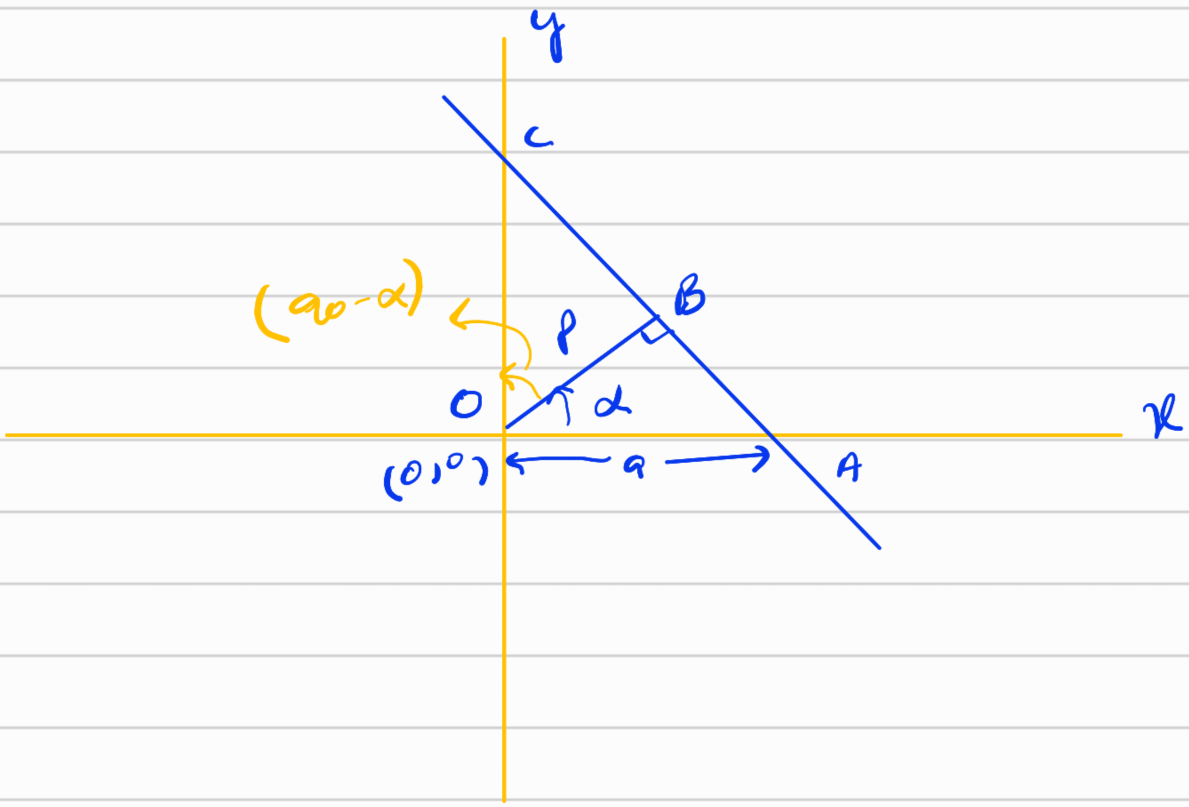
$$\Rightarrow ya = -bx + ab$$

$$\Rightarrow bx + ay = ab$$

$$\Rightarrow \frac{bx}{ab} + \frac{ay}{ab} = 1$$

$$\Rightarrow \boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

Normal Form



In $\triangle OBA$

$$\frac{a}{p} = \sec\alpha$$

$$a = p\sec\alpha$$

In $\triangle OBC$

$$\frac{b}{p} = \sec(90-\alpha)$$

$$\Rightarrow b = p\,\mathrm{cosec}\,\alpha$$

Eq. of straight line (AC)

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{p\sec\alpha} + \frac{y}{p\,\mathrm{cosec}\,\alpha} = 1$$

$$\Rightarrow \frac{x\cos\alpha}{p} + \frac{y\sin\alpha}{p} = 1$$

$$\Rightarrow \boxed{x\cos\alpha + y\sin\alpha = p}$$